Physiology

for 1st year medical students

Sensory

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❖ Anatomical Organization of the nervous system:

1- Central Nervous System (CNS): brain and spinal cord.

- contains centers for integration of inputs and for production of output signals.
- ***three major levels:***
  - **a. Spinal cord level:** for immediate automatic activities as withdrawal reflexes and evacuation of bladder and rectum.
  - **b. Lower Brain Level:** control of subconscious activities as CVS and respiratory functions, maintenance of posture and equilibrium ……..
  - **c. Higher Brain or Cortical Level:** control of motor and sensory functions and others as: thinking, learning, memory and speech.

2- Peripheral Nervous System (PNS): subdivided into:

- **a. Sensory Division: (Afferent)**
- **b. Motor Division: (Efferent):** further subdivided into:
  - **Somatic:** controls voluntary movements as skeletal muscles.
  - **Autonomic**: controls involuntary functions as viscera, smooth muscles, cardiac muscles and glands.

Synaptic Transmission

- **Definition**: transmission of impulse (action potential) from one neuron to another along a synapse.
- **Synapse**: junction between 2 neurons: **axon terminal** of one neuron **(presynaptic neuron)** and a second neuron **(post synaptic neuron).**

❖ types of synaptic transmission:

1- Electric: neurons are communicated by **low-resistance gap-channel pathway** → passage of electric current directly.

N.B.: rare in humans and is common in invertebrates.

2- Chemical: release of a chemical substance (neurotransmitter)**.**

➢ *Functions of Synapses:*

a. Facilitating transmission.
b. Blocking transmission.
c. Amplification of impulses.
d. Distribution of impulses.
e. Changing one impulse to repetitive impulses.
f. Integration between different types of neurons.
g. Interpretation, processing and Storage of information.

N.B. synapses are ***dynamic structures:*** increasing and decreasing in complexity and number with use and experience.

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➢ ***Functional Anatomy of a Synapse***

1) Presynaptic Terminal: (or synaptic knob): contains:

a. ***mitochondria*:** provides ATP for synthesis and exocytosis of transmitters.

b. A large number of synaptic ***vesicles*** contain synaptic transmitters. ***three types*:**

1. **Small clear** for rapidly acting transmitters e.g. acetyl choline, glycine, glutamate and GABA.
2. **Small granular** containing catecholamines.
3. **Large granular** contain slowly acting neuropeptides.

***(2) synaptic cleft*:** space containing **ECF** rich in Na^+, Cl^- poor in K^+.

(3) postsynaptic membrane (PSM)*:** contains ***Receptors for neurotransmitters. types**:**

a. Ionotropic receptors: part of **Ligand-gated ion channels:** affect neuronal activity **directly**

b. Metabotropric receptors: (mostly G protein coupled receptors) affect activity neuron **indirectly** by **second messenger** e.g. **cAMP: slower** and **more prolonged** effects, as:

a. Open or close specific ion **channels.**
b. Activate **enzymes** e.g. protein kinases: phosphorylation of proteins.
c. Activate **gene** transcription: formation of new proteins.

Mechanism of Synaptic Transmission

1- Release of chemical transmitter from presynaptic nerve:

- Arrival of action potential opens **voltage-gated Ca^{++}** channels
- Ca^{++} enters knob according to concentration and electric gradients.
- Influx of Ca^{++} triggers **binding** of SNARE proteins:
  - V- SNARE (Synaptobrevin) on synaptic vesicle.
  - T- SNARE (Syntaxin) on presynaptic membrane
- **exocytosis** of chemical transmitter into the synaptic cleft.
- N.B.: amount of transmitter is proportional to amount of Ca^{++} entered.

2- Binding of Transmitter with its Receptors: changes permeability to ions.

3- Generation of Postsynaptic Potential: Change in ion fluxes leads to change in its membrane potential → **may become:**

a. less negative = excitatory postsynaptic potential. **(EPSP).**
b. more negative = inhibitory postsynaptic potential. **(IPSP).**

magnitude and duration of EPSP and IPSP depends on amount of released Neurotransmitter. "not All or None".

4- Removal of Neurotransmitters from the Synaptic Cleft:

a. ***Passive diffusion*** away from synaptic cleft
b. ***Inactivation*** by specific enzymes within synaptic cleft, e.g. acetyl choline
c. ***Active reuptake*** into axon terminals, e.g. catecholamines, dopamine, serotonin.
d. Removal by ***glial*** cells.

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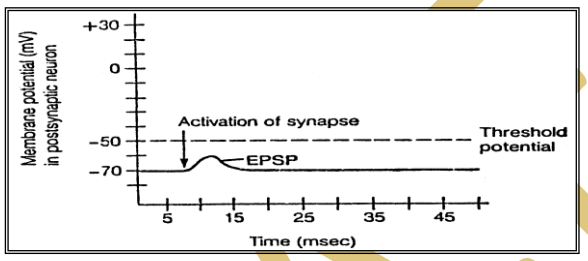
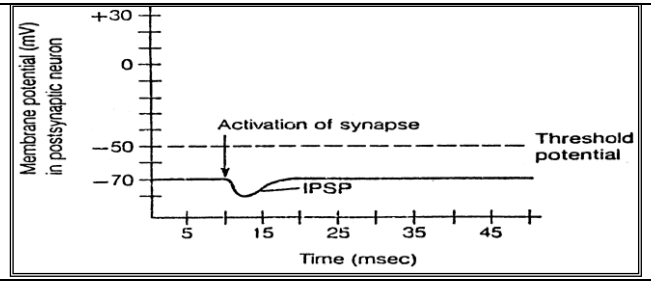
Types of Synaptic Potentials

Postsynaptic potentials

	1. Excitatory postsynaptic potential [EPSP]:	2. Inhibitory Postsynaptic Potential [IPSP]
Definition	local state of partial depolarization	local state of hyperpolarization
Ionic basis	• **excitatory** neurotransmitter binding to receptor → opening of ligand gated Na^+(or Ca^{++}) channels • inside of cell membrane **less negative** → nearer to firing level → more excitable. • membrane ***facilitated*** i.e. needs a weaker stimulus to be excited than at rest.	• **inhibitory** neurotransmitter bind with receptor → opening of ligand gated K^+ or Cl^-channels or closure of Na^+ (or Ca^{++}) channels. • inside of the cell membrane **more negative** → away from firing level (***threshold)*** → less excitable. • membrane is ***inhibited*** i.e. needs a stronger stimulus.
amplitude	1- 5 mv	
Duration	reaches a maximum after 1-5 msec then gradually declines	
N.B.:	**does not obey all or none rule = graded:** Magnitude and duration depends on amount of neurotransmitter	
		

❖ ***Summation*:** EPSPs have to be summated to reach threshold. ***Types***

1- **Temporal (Time):** **rapid repetitive** stimuli originating from **ONE** presynaptic neuron

2- **Spatial (Space):** multiple **simultaneous** stimuli from **MANY** presynaptic neurons

N.B.: IPSP can be **summated**

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3. Grand Post-Synaptic Potential (GPSP): sum of all EPSPs and IPSPs (neuron receives thousands of presynaptic inputs)

❖ ***possible outcomes:***

1. **EPSPs = IPSPs** → postsynaptic membrane potential will remain at ***resting level.***
2. **EPSPs < IPSPs** → postsynaptic away from threshold and membrane **inhibited**
3. **EPSPs > IPSPs** → postsynaptic neuron doesn't reach threshold →membrane is facilitated or in an ***excitatory state***.
4. **EPSPs > IPSPs** → postsynaptic neuron reach threshold → ***action potential***.

Example: micturition reflex is influenced by many signals but the response will vary:

a- Sensory signals from receptors in urinary bladder.
b- Facilitatory and Inhibitory signals from brain stem.
c- Voluntary control from the cerebral cortex.

Presynaptic Potentials

- release of NT from **presynaptic** affected by a **third neuron**.
- N.B.: **3rd neuron** synapses axoaxonically with excitatory presynaptic neuron **before** it reaches the synapse.

	Presynaptic Inhibition	**Presynaptic Facilitation**
THIRD NEURON	**inhibitory** releases an inhibitory transmitter (e.g **GABA**), which either **opens Cl^-** or **K^+ channels**.	excitatory neuron releases *"serotonin"* → increases **cAMP** in the presynaptic terminals → phosphorylates **K^+ channels and closes** them.
presynaptic membrane	**hyperpolarize → reduced Ca^{++}** influx → **reduced release of NT** and transmission from the presynaptic to the postsynaptic neuron	delays repolarization = **prolongation of depolarization**: voltage gated **Ca^{++} channels opened** for a longer period → Ca^{++} influx → **increases release of NT**
Example	**pain inhibiton**	**Sensitization of memory**
significance	↑ neurotransmitter release from presynaptic	Selective block of synapse without affecting postsynaptic neuron excitability

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➢ Initiation of Action Potentials at the Axon Hillock: -

- Few Voltage gated Na+ channels are present in cell body and dendrites, but numerous in axon hillock
- EPSPs occurring on cell body or dendrites spread electrotonically (by current sink) → axon hillock: initiating an action potential

Factors affecting synaptic transmission

1- Changes in internal environment:

a. pH:

- **Alkalosis: increases** neurons excitability & facilitates transmission: decrease ionized Ca^{++}. (may lead to convulsions: if epileptic patient hyperventilates, this may precipitate a fit)
- **Acidosis: decreases** synaptic transmission due to increase ionized Ca^{++} (may lead to coma, e.g. in diabetic Ketoacidosis).

b. Hypoxia: inhibits synaptic transmission: interruption of cerebral circulation for 3-5 seconds may cause unconsciousness & prolonged ischemia for minutes cause brain damage.

c. Hypoglycemia: inhibits synaptic transmission (Glucose is the main fuel of brain)

d. Hypocalcemia: facilities transmission as it increases excitability of the post synaptic membrane (leading to tetanic convulsions)

2- Drugs: Most of the drugs act on the CNS alter synaptic mechanism, e.g.:

a. Theophylline, theobromine, and caffeine ↑ transmission by depolarizing the postsynaptic membrane.

b. Analgesics, hypnotics and anaesthetics: ↓ transmission = **stabilizing** membrane causing hyperpolarization **or** by interference with synthesis, release or reuptake of certain neurotransmitters.

c. Strychnine: ↑↑↑↑ transmission: competes with inhibitory transmitters leaving **excitatory** pathways unaffected: convulsions.

3- Diseases:

Tetanus toxin	*Botulism toxin*
prevents release of inhibitory GABA (or glycine)	blocks release of excitatory Acetyl Choline
spastic paralysis: spasm occurs especially in jaw muscles → lock jaw then affects respiratory muscles →death.	**flaccid paralysis**

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Characters of Synaptic Transmission

I- Forward (uni) direction: from presynaptic to postsynaptic neurons (neurotransmitters released from presynaptic).

II- Synaptic delay: time taken by an impulse to be conducted through the synapse. It equals about **0.5 m. sec**: so we can calculate number of synapses

III- Fatigue: decreased impulses discharged after repetitive stimulation. **due to:**

1. **Exhaustion** of synaptic vesicles in the presynaptic terminals.
2. **Inactivation** of postsynaptic receptors.

- **Benefit: stabilize** the nervous system e.g. stopping discharge from overexcited areas in CNS during epileptic seizures.

IV- Synaptic plasticity: ability to change function according to demand (synaptic transmission strengthened or weekened)

❖ represent forms of learning and memory.

❖ **Forms of Synaptic Plasticity**

	Post tetanic potentiation (PTP):	**Habituation**	**Sensitization**
Definition	Continuous discharge from postsynaptic neuron after brief tetanizing stimuli by presynaptic neuron	Gradual loss of response to repeated benign stimuli	Augmented response of postsynaptic to benign stimuli if coupled with a noxious one.
Duration	**few seconds to minutes**	**Up to 3 weeks**	
Ionic basis	repetitive stimulation → accumulation of Ca^{2+} in presynaptic neuron → more release of neurotransmitter	gradual inactivation of Ca^{2+} channels in presynaptic → ↓Ca^{2+} → ↓ release of neurotransmitter	(presynaptic facilitation): ……..
Importance	mechanism of immediate memory as telephone number (7-10 digits)	Neglecting unimportant stimuli e.g. noises and focusing on important ones	1. In memory **2.** defensive response to avoid injury

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	Long term potentiation (LTP):	**Long term depression (LTD)**
Definition	Persistent increase in synaptic strength after strong intensity high frequency stimulation	Persistent decrease in synaptic strength after weak intensity low frequency stimulation
mechanism	• **Glutamate** from presynaptic neuron binds to AMPA and NMDA receptors in postsynaptic membrane. • Activation of AMPA causes Na^{+} inflow: depolarizes postsynaptic membrane and relieves Mg^{++} block in NMDA receptor channel, Ca^{++} enters neuron.	
	• **High intensity & frequency stimuli:** rapid increase in cytoplasmic Ca^{++} activates (Calmodulin Kinase II): moves more AMPA receptors into postsynaptic membrane.	• **Low intensity & frequency stimuli** slow increase in cytoplasmic Ca^{++} activates (Calcineurin phosphatase I): removal of AMPA from postsynaptic membrane
Importance	in various parts of CNS, specifically in hippocampus contributes to memory storage → learning & long term memory	memory decay.

N.B.: Memory classification:

1. **Short term:** seconds to few minutes
2. **Intermediate:** up to 3 weeks
3. **Long term:** days, months or even lifetime: requires structural changes in synapse as protein synthesis

Memory consolidation: transformation of short term memory into long term by hippocampus to be stored in neocortex

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Chemical Transmitters (Neuro-Transmitters)

- endogenous chemical substances released from presynaptic neurons that transmit signals between neurons across a synapse
- may act as a **neurotransmitter** in brain & as a **hormone** elsewhere

❖ **Types: More than 100** in two different groups:

	Small molecule, rapidly acting transmitters	**Large molecules, slowly acting neuropeptides**
synthesized in	presynaptic **knobs**	in **soma**, then transported to the terminal knobs
stored in	**small** vesicles	**large granular** vesicles
vesicles	**recycled** and reused	**autolyzed** and not reused
released	**rapidly** in **large amounts**	**slower** in **smaller amounts** (but **more potent)**
receptors	**ionotropic** leading to rapid activation of ion channels	**metabotropic receptors** leading to activation of 2nd messenger system
responses	most of the **acute actions** e.g. transmission of sensory signals and motor signals to muscles	**more prolonged actions** as, long term opening or closure of ion channels, changes in number of receptors or genes
types	a. **Acetyl choline** b. **Amines:** Norepinephrine, Dopamine, Serotonin c. **Amino acid Neurotransmitters:** • Excitatory: Glutamate and Aspartate • Inhibitory: GABA and Glycine d. **Gases:** Nitrous oxide (NO)	a) **Hypothalamic releasing peptides**: TRH, CRH. b) **Pituitary peptides**: GH, TSH, ADH c) **Opioid peptides**: endorphins, enkephalins. d) **GIT peptides**: CCK, VIP, gastrin, secretin. e) **Other peptides**: substance P, Angiotensin II, Neuropeptide Y, ANP, BNP. f) **Purines:** adenosine.

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Organization of Neurons in Neuronal Pools

- **neuronal pool:** collection of neurons carrying the same function in CNS
- ***organizations are:***

	I. Divergence	*II.* Convergence
Definition	**ONE** neuron stimulates **MANY** neurons	**MANY** neurons stimulate **ONE** neuron
Functions	**a- in same direction:** allows **Amplification** of signal, e.g. one motor cell in cortex supply many muscle fibers. **b- in different directions:** allows **Distribution**, e.g. painful stimulus: **a)** stimulates AHCs of muscles on both sides **b)** ascends to brain stem & sensory cortex	***a.*** **SUMMATION** of input signals: **intensification** of subthreshold signals to threshold to discharge. b. **from different sources**: **integration** of signals e.g. α motor neuron receives different excitatory and inhibitory inputs

III. **Excitation field:** neurons synapsing one afferent neuron.

a) Discharge zone : Neurons in **center** receive large numbers of knobs: reach threshold and **discharge** nerve impulses.

b) Facilitated zone(subliminal fringe): Neurons in **periphery** receive few numbers of knobs: **only facilitated**

N.B.: area of discharge zone and subliminal fringe depends on strength of stimulus: the **stronger** the **wider** the **discharge** zone

❖ Facilitation phenomenon:

overlap	excitation fields of two inputs **overlap at facilitation zones (subliminal fringe)**,
outcome (if stimulated simultaneously)	**more than the sum** of outcome when each neuron is stimulated separately
function	2 neighboring neurons facilitate each other
example	**Referred pain**

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IV. Inhibitory Circuits: for **stabilizing** nervous system by preventing excessive spread of signals.

	1) Lateral inhibition	*2)* Negative feedback inhibition	*3)* Reciprocal innervation
Definition	only central neuron is stimulated while neurons at the periphery are inhibited by inhibitory interneuron.	When a motor neuron stimulated it gives off recurrent collateral that synapse with **Inhibitory Interneuron (Renshaw cell)** which terminates on motor neuron itself	**stimulation of one muscle** and **inhibition of its antagonist** by **one nerve**. (through inhibitory interneurons)
Function	**Sharpening of sensation:** prevents blurring effect of the fringe fibers e.g. 1. Horizontal cells in retina inhibit peripheral bipolar cells ***2.*** Tactile localization in skin	1) Allows focusing and sharpening. 2) Dampens activity of motor neurons and prevents over Excitation	contracted muscle carry its function unopposed eg. A painful stimulus **stimulates flexors** and **inhibits extensors**

V. **Activating Circuits: (After Discharge): Continued output** discharge after stoppage of stimulation of input.:

1) Parallel circuits	2) Reverberatory (Oscillatory = closed circuits):
• excitatory input signal reaches output through many circuits of different number of interneurons. • Synaptic delay leads to arrival of successive impulses to output. • ***Function:*** prolongation of signal discharge after input signal is over • ***Example:*** while looking at image , we don't notice when we blink	• output neuron sends collaterals to restimulate itself (positive feedback) • May continue for a long period or even throughout whole life • base of **wakefulness and sleep** and other tonically discharging centers, e.g. respiratory centers. • **Duration determined by:** 1) Number of synapses in the circuit 2) Fatigue of synapses. 3) Facilitatory or inhibitory impulses entering the circuit.

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SENSORY NERVOUS SYSTEM

- **Sensation:** conscious awareness of stimulus (Information) by sensory system
- **Perception:** **understanding** of the sensation's meaning**.**

Sensory Receptors

- **Definitions:**
  - **Sensory receptor:** specialized (modified) structure at peripheral termination of afferent nerve fibers.
  - **Sensory unit:** single sensory afferent with all its receptor endings.
  - **Receptive field of a neuron:** region which contains all the receptor endings of a sensory neuron (stimulus produces a response in that afferent neuron).

- **Functions:**

  1- Detectors: **detect stimuli** (changes in the internal or external environment).

  2- Transducers: transform energy in stimulus (chemical, mechanical, etc) into **electrical energy (receptor potential).**

  3- Generators: generation of **action potential** (nerve impulses) in the sensory nerves

N.B.: Signal Transduction: process by which a stimulus is transformed into an electrical response

- **Properties of receptors:**

[1] Specificity:

- Each type of receptor is highly sensitive to one specific type of stimulus = **adequate stimulus** "e.g. light, sound, pressure, etc.".
- Receptors **respond to** other stimuli but:

  1) **much higher strength**

  2) **sensation** perceived is that of adequate stimulus.

  e.g. **rods and cones** in retina can **respond** to a strong **mechanical** stimulus, but give sensation of light

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[2] Excitability:

- When stimulated, a receptor generates **"Receptor (or Generator) Potential"**: **graded electric response:**
  (as an example: Paccinian corpuscle was chosen . Pressing on capsule by a rod leads to development of receptor potential)**.**

❖ ***Transduction of Sensory Signals = (Mechanism of Receptor Potential = Generator Potential)***

1- Stimulation of the receptor:

a. stimulus causes **opening of ion channels** "transduction channels" → **Na^+ or Ca^{++} inflow**.

b. number of opened channels is proportional to intensity of the stimulus

2- Creation of Receptor Potential: Na^+ or Ca^{++} influx creates **local partial depolarization**

3- Electrotonic spread of receptor potential: local circuit of current (current sink) **spreads passively** to adjacent part of sensory nerve or 1st node of Ranvier (myelinated neurons) = action potential (A.P.) generating zone (it has low firing level = many **voltage gated Na^+ channels)**

4- Generation of Action Potential: If amplitude of receptor potential is strong to depolarize [A.P. generating zone] to threshold, it opens **voltage gated Na^+ channels** and an **action potential** is produced & propagated along the sensory nerve to CNS.

❖ **Properties of Receptor Potential (local response):**

1. **non-propagated**
2. does **not obey all or none-law**, i.e. can be graded.
3. **no absolute refractory period**, i.e. can be summated.
4. **duration (5-10 msec.)** is longer than action potential duration (2 msec.) so can cause repeated action potentials.
5. **maximum amplitude is 100 mV** (at maximum permeability to Na+ ions).

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❖ **Relationship between strength of stimulus & magnitude of receptor potential:** **As strength of stimulus increases, magnitude of the R.P. increases.**

- **N.B.:**
  - amplitude increases rapidly at first then less rapidly at high stimulus strengths.
  - allows receptor to be sensitive (have an extreme range of response) to very weak sensory stimuli without reaching maximum firing until sensory experience is extreme
- **Relation of the receptor potential magnitude to number of impulses generated:**
  - The **more** the receptor **potential** rises above threshold level, the **greater** the **frequency** of the action potential

[3] Adaptation of receptors:

- **Definition:** ***gradual*** **decline** of **receptor potential** and **frequency of action potentials** generated in spite of the continued presence of stimulus at ***constant*** strength.
- **Causes:** Each receptor has its **own** property; adaptation of **mechanoreceptors** occurs by:
  a. **Readjustment and remolding** in the structure.
  b. **Accommodation** of terminal nerve fiber or 1st node of Ranvier to the stimulus, due to inactivation of Na^+ channels.

❖ **receptors are classified into:**

Rapidly adapting (phasic)	Slowly adapting (Tonic)
adapt rapidly	either slowly adapting or they do not adapt at all
• **discharge strongly when a change is taking place** • **response** when the stimulus is **"ON"** and another **response** when the stimulus is **"OFF"**	discharge as long as the stimulus is applied
detect movement, rate of movement and vibration e.g. **touch** and **pressure receptors** (Meissner's & Paccinian corpuscles).	• Keep the brain continuously informed e.g. **pain receptors**, **muscle spindles**and **arterial baroreceptors**.

N.B. ***Some receptors have both phasic and a tonic component, e.g. proprioceptors and thermoreceptors***

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Sensory Code

Definition: ability of **CNS** to recognize **modality** (type), **locality** and **intensity** of sensation. (All stimuli produce action potentials)

A) Modality of sensation: depends on:

1. **Adequate Stimulus:** Each receptor is highly sensitive to a particular type of **stimulus.**
2. **Muller's law of specific nerve energy:** Whatever method of stimulation, **sensation** given is that the receptor is specialized to.
3. **Labeled line principle:** Each sensation reaches CNS in a **specific pathway:** Stimulation of this pathway at any point by any form evokes its specific sensation.

B) Locality of the stimulus:

- **Law of Projection:** There is a separate **representation area** for each part of the body in the cerebral cortex, (i.e. point to point representation). When an impulse reaches the specific area in the cortex, it **"projects"** this stimulus to its original site.
- **Phantom limb: a** phenomenon in which pressure on the nerves in the stump of an amputated limb makes the patient feel the pain as though it is coming from the absent limb

C) Intensity of sensation: It is encoded by change in:

1- **Number of receptors activated**: Higher intensity stimuli will recruit more receptors ["**recruitment**" of receptors]

2- **Frequency of impulses**: Increasing stimulus intensity increase frequency of impulses with increased intensity of sensation felt.

Somatic Sensations

- Sensation: feeling produced by **change in environment** or by application of a **stimulus** to **receptors** or nervous **pathway**.
- **Sensations are either:**
  - **a) Special senses**
  - **b) General sensations** classified into:
    - 1- **Somatic sensations** from different parts of the body.
    - 2- **Organic sensations** as *thirst, hunger, and sexual sensations.*
  - **c) Emotional as**: *fear, sadness and so on.*

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Somatic Sensory afferents

- Peripheral branches of dorsal root ganglion or sensory fibers of cranial nerves. **Classifications:**

1. According to diameter: (from largest): I II III IV **2. According to velocity: (from fastest) A B C**

N.B.: type of afferent is determined by importance on information transfer: position by A alfa while slow pain by C

Somatic Sensory Pathways

	Dorsal Column	**Spinothalamic (Ventrolateral) Tract**
	Unimodality: mechanoreceptive sensations	**polymodality** : different types of sensations
Receptors	mechanoreceptors for: a. Fine touch and pressure. b. Vibration. c. Stereognosis d. Position	a. Pain b. Temperature c. Crude touch and pressure d. Tickle and itch
Afferents	myelinated Aα and Aβ fibers	thin myelinated type Aδ and unmyelinated type C fibers
1st Order Neurons	dorsal root ganglia (DRG).	
	fibers then ascend upwards on ipsilateral (same) side as dorsal column (**gracile and cuneate tracts**)	
2nd Order Neurons	gracile and cuneate nuclei in medulla on same side; Fibers then cross to opposite side	ipsilateral (same side) posterior horn cells of spinal cord then cross to opposite side: **a.** Fibers of **pain and temperature** cross immediately infront of central canal, and ascend as **lateral spinothalamic tract** **b.** Fibers of **crude touch, tickle and itch** cross farther from central canal and ascend as **ventral spinothalamic tract**
3rd Neurons	**posteroventral (PVN) = ventrobasal (VBN) nuclei of = thalamus**: to somatic sensory cortex on opposite side	

N.B.: passing brain stem, both pathway are joined by additional fibers from nuclei of trigeminal nerve carrying sensations from head.

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❖ Role of the thalamus and cortex in sensory perception

a. All sensory tracts, **except the olfactory**, synapse in the thalamus on their way to the cerebral cortex.
b. subject is **crudely aware** of sensation but without details: **Fine gradations** and **sensory localization** at level of the cortex.
c. Pain perception mainly occurs in thalamus.

Somatic Sensory Cortex

	Somatic sensory area I (SSI): "area 3, 1, 2"	**Somatic sensory area II: (SSII):** area 40	**Somatic Association area** areas 5,7
Site	post central gyrus in pariatal cortex	, below and behind SSI	,posterior to SSI
Body representation	map of body known as **spatial orientation**. 1) **crossed**: from opposite side of body. 2) **inverted**: upside-down except face. 3) area **directly proportional** to number of receptors (thumb & lips have large area while back and trunk have smaller areas)	not as detailed or as complete as SI. Face anteriorly, arms centrally and legs posteriorly	
Connections		receives sensory signals from both sides of body & from SSI	receives signals from: 1. S.S. I & II. 2. VBN of thalamus. 3. Visual cortex 4. Auditory cortex
Functions	**discriminative**: 1) Recognition of fine touch, stereognosis, vibration and proprioceptive sensations. 2) Localization of source of sensation 3) Discrimination of fine gradations		Integration and interpretation of sensory inputs to understand their meaning

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Types of Somatic Sensations

I- Mechanoceptive Sensations

	Superficial sensations: from skin & subcutaneous tissue		**Deep sensations:** from muscles & joints			
	Crude Touch	**crude Pressure**	**fine pressure**	**Vibration**	**Position**	
					Static	**Kinetic**
Definition	Touch with poor discrimination e.g. feeling of clothes.	feeling of mechanical pressure	discriminate between different weights	feel vibrating stimuli (more magnified on bony prominences)	position of different parts of body (angle of joint)	rate of movement
Receptors	Free nerve endings & Hair end organ	Pacinian corpuscles and spray type endings		- Paccinian 500 c/sec. - Meissner 80 cycle/sec.	**Slowly adapting:** a. Muscle spindles in skeletal muscle. b. Golgi tendon organs in tendons.	**Rapidly adapting** as Pacinian corpuscles
Afferent	**A δ**		**A_B**		**A alfa**	
Pathway	**ventral spinothalamic tract**		**Dorsal column**			
examination	piece of cotton passed on skin	Feeling weights on hand	different weights on hand	(tuning fork) on bony prominences	Identify limb position	

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Cortical sensations: require cortical integration					
	Stereognosis	**graphathesia**	**Tactile localization**	**Two point discrimination**	**Texture**
Definition	ability to recognize familiar objects in hand with eyes closed.	ability to recognize symbols traced on skin	ability to localize site of tactile stimulus on the skin	ability to recognize two touched points as two separate points. measured by a threshold distance: minimal distance below which the two points are felt as one **best at** lips and fingertips (shortest distance) and **worst at** the back (largest distance)	ability to differentiate between texture of different materials e.g. silk or wool
Receptors	mixture of tactile receptors.		Meissner's & Merkels		
Afferent	A_B				
Pathway	**Dorsal column**				
examination	familiar objects placed in hand	Drawing numbers in the palm	Touch skin with blunt object	Using caliber	Touch with different materials

❖ **Clinically**:

- **Impaired stereognosis** is an early sign of damage in the cerebral cortex (SSI & association area)
- **Impaired vibration sense**: early diagnostic sign in diseases causing degeneration of posterior column such as pernicious anemia

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II- Thermoceptive temperature Sensations

❖ **Receptors & Afferents & Pathway:** Free nerve endings → C fibers (warm) and A^{δ} fibers (cold) → Lateral Spinothalamic tract

❖ **Types of thermoreceptors:**

Cold pain receptors	Cold receptors	Warm receptors	Warm pain receptors
$<10^{o}C$	$10 - 40^{o}C$	$30 – 45^{o}C$	$>45\ ^{o}C$

- At zero ^{o}C, no receptors discharge = anaesthesia =.
- between $30^{o}C$ and $40^{o}C$: both cold and warm receptors discharge and are very sensitive to fine gradations

❖ **Distribution of thermal receptors:**

1. Located in and respond to temperature of the **subcutaneous tissue** (not to the environmental temperature).
2. **numerous in** lips & finger tips and **least in** the trunk
3. **Cold** receptors are **4-10 times** more than warm receptors
4. **widely separated**: wide area of skin has to be exposed to allow spatial summation.

❖ **Adaptation of thermal receptors:**

a. **between** skin temperature **20-40** ^{o}C: warm receptors adapt faster than cold receptors.
b. **2 components**: rapidly at first (phasic component) but continue to generate impulses at low frequency (tonic component).

❖ **Mechanism of stimulation:**

1- **chemically** by changes in concentration of metabolites due to change in metabolic rate (each **10** ^{o}C increase, doubles concentration).

2- membrane ion channels: **"Temperature-activated Transient Receptor Potential" (TRP)** ion channels

3- respond markedly to **changing temperatures** rather than steady states:

- Warm receptors increase their firing when temperature is increasing.
- Cold receptors increase their firing when temperature is decreasing.

4- **relative** Sensation: **depends** upon **original skin temperature**. ($35^{o}C$ feel warm if skin is $30^{o}C$ and feel cold if skin is at $40^{o}C$)

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III- Pain Sensation

- **Pain**: unpleasant protective sensory and emotional experience if there is physical or potential tissue damage.

➢ **Pain Receptors (Nociceptors):** Free Nerve Endings

❖ **types:**

1- **Mechanical**: stimulated by mechanical injurious stimuli, e.g. cuts, bruises.
2- **Thermal:** respond to extremes of temperature and capsaicin (substance present in hot chilli).
3- **Chemical:** stimulated by chemical injurious elements.
4- **Polymodal:** respond to all stimuli.

❖ **Distribution of Pain receptors:**

- **most numerous** in superficial layers of the **skin**, periosteum, peritoneum, pleura, joints, arterial walls, dura and tentorium.
- **less distributed** in deep tissues and very few in internal viscera.
- **absent** in liver parenchyma, lung alveoli, and brain tissue (pain insensitive structures).

❖ **Adaptation:** slowly or non-adaptive receptors.

❖ **Pain Transduction by Nociceptors:**

- Noxious stimuli open **transduction ion channels** in pain receptors: increase membrane permeability to Na+ &/or Ca++ :

a) Mechanical stimuli open "degenerin" ion channels.
b) Thermal stimuli open "temperature sensing TRP" ion channels.
c) Chemical stimuli open "acid sensing" ion channels.

❖ **Pain Sensitizers (OR Chemical mediators of pain):**

- **damaged tissues** release a number of **pain** and **inflammation** producing **chemical** substances into ECF. **include**: histamine, serotonin, K+, substance P, ATP, bradykinin and prostaglandins.
- **Prostaglandins** further **sensitize** nociceptors, lowering their pain threshold, and producing **primary hyperalgesia**.

N.B: Salicylates and other non steroidal anti-inflammatory analgesics (NSAID) reduce pain by inhibiting prostaglandin synthesis.

➢ **Threshold of Pain:** the **same** for all people but **reaction** to pain **differs:** Pain starts if **skin temperature reaches 45°C**

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➢ **Types of pain sensation:**

I- According to site: a. Cutaneous. b. Deep. c. Visceral. D. neuropathic

II- According to quality (Single painful stimuli may evoke both):

	Fast Pain	Slow pain
Quality	Sharp, pricking, acute	Dull aching, Burning, throbbing
Onset	Immediate	Delayed
Duration	Short duration	Long duration & increase with time
Localization	**Well localized**	**Poorly localized (diffuse)**
Stimulated Receptor	Mechanical & thermal	All types of pain receptors
Felt in	Skin and Parietal surfaces	All tissues (Skin, deep tissues, viscera)
Carried by (afferents)	**Aδ fibers**	**C fibers**
Blocked mainly by	Hypoxia & compression	Local anaesthesia (e.g. cocaine)
Neurotransmitter	**Glutamate**	**Substance P**
Pathway	Neo-spinothalamic Tract	Paleo-spinothalamic Tract
Relay of 2nd order neurons	VBN of thalamus	Reticular Formation → intralaminar thalamic N
Termination of fibers	sensory cortex	whole cortex
Pain perception	Thalamus and Cortex	Mainly In Thalamus
Motor reflexes	Flexor Withdrawal reflex	Guarding: hypertonia of overlying muscles
Autonomic reactions	Pressor response: (↑BP & HR)	Depressor response: (↓BP & HR) & Nausea and vomiting
Emotional reactions	Anxiety	Depression

N.B. The two types of pain can be dissociated by:

1. Moderate **compression** of a nerve trunk **block Aδ-fibers:** fast pain is lost while the slow type remains.
2. Low concentrations of **local anaesthetic blocks C-fibers:** slow pain is lost while the fast type remains.

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- **Pain Pathway:**

	neospinothalamic tract (**Fast pain**)	**paleospinothalamic** tract (**Slow pain**)
First order neurons	dorsal root ganglia **(DRG)**	
Second order neurons	Posterior horn cells (PHCs) on the same side	
	Aδ fibers release **glutamate** and synapse with PHCs in lamina I and V	C fibers release **Substance P** and synapse with PHCs in lamina II and III (i.e. substantia gelatinosa of Rolando or **SGR**).
	Pain fibers then cross and ascend in **lateral spinothalamic tract** in **opposite** side of spinal cord.	
Third order neurons	end in ventrobasal nuclei of thalamus **(VBN),** then to **somatic sensory cortex**	reach the brain stem, where: a) **10%:** join the neospinothalamic tract to **VBN** of thalamus then end in somatic sensory cortex. b) **90%:** to **reticular formation**, then **intralaminar (non specific) nuclei of thalamus** then activate **whole cortex**

- **Reaction to Pain: pain fibers give tributaries to**:
  a. **Anterior horn cells** in the spinal cord for accompanying motor reflexes.
  b. **Reticular Activating System** for cortical activation and arousal reactions
  c. **Hypothalamus** for accompanying autonomic reactions.
  d. **Limbic system** for emotional reactions.
  - **Arousal reaction to pain signal:** intralaminar (non-specific) nuclei of the thalamus and reticular formation of the brain stem have a strong arousal effect on nervous activities through the **reticular activating system (RAS)**
- **Function of cerebral cortex in Perception** (conscious awareness) **of pain signals:**
  a. **Localization** of source of pain: fast pain is very localized.
  b. **Discrimination** of pain modality and intensity.
  c. Interpretation of its **meaning**.
  d. **prefrontal lobe** of the cortex contributes to **emotional** and behavioral effects of pain

- **N.B.: Pain perception** can occur subcortically in the **Thalamus** : cortex **not essential**

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Types of Pain Sensation

	A] Cutaneous Pain	B] Deep (musculoskeletal) Pain	Visceral Pain
	in **skin**	muscles, tendons, ligaments, joints, bones	from internal viscera
type	starts **fast sharp** pain **followed** by **slow dull pain**	**Slow**: conducted along thin C fibers	• **Slow**: along **C fibers** • **Fast (Aδ fibers): in parietal walls** (pleura, peritoneum and pericardium) • **N.B.:** if disease spreads to wall, it causes sharp well localized pain • Afferent reach CNS via autonomic nerves.
Causes		1. **Trauma**: e.g. broken bone. 2. **Inflammation** e.g arthritis. 3. **Muscle spasm**: e.g. injury → reflex contraction of near muscle → ischemia → more pain → spasm → ischemia → pain …etc (vicious circle). 4. **Ischemia**: • thrombosis or mechanical pressure. • Cause: **accumulation of metabolites and proteolytic enzymes**. • Pain **aggravated** by increased **activity** and **relieved** by **rest**. • **Example:** Anginal pain and intermittent claudication in skeletal muscles.	• Viscera have **few pain receptors:** localized cut injury does not cause pain, diffuse stimulation of a large area cause pain. 1- **Inflammation:** inflammatory products eg: appendicitis. 2- **Ischemia:** accumulation of metabolites as H^+, K^+, bradykinin or prostaglandin. 3- **Spasm of a hollow viscus:** a. Mechanical stimulation of pain receptors. b. Obliteration of vessels causing ischemia. 4- **Overdistention of hollow viscus:** mechanical and ischemic stimulation. 5- **Chemical irritation** e.g. peptic ulcer 6- Compression or infiltration by **tumors.**

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Characters	**accurately localized**, due to: 1. high density of receptors in skin. 2. fast pain fibers reach cortex. 3. touch and vision help.		1. **Dull aching pain** 2. **Diffuse** (Poorly localized): few receptors and less representation in cortex. 3. **Depressor (autonomic)**: hypotension, bradycardia, nausea, vomiting 4. **Guarding**: reflex spasm of skeletal muscles over affected area. 5. Usually **referred** to surface area.

❖ **Neuropathic Pain:**

- chronic type of pain due to damage to or pathological changes in nerve fibers in peripheral or central nervous system.
- **Characters:**
  - a. burning, electric, tingling or shooting pain.
  - b. bouts or paroxysms.
  - c. accompanied by hyperalgesia and/or parasthesia.
- **Examples:**

1. Trigeminal neuralgia: (tic douloureux): unilateral facial pain in areas supplied by 2nd and 3rd divisions of the trigeminal nerve.

2. **Herpes Zoster:**
   - **viral** infection: herpes virus attacks a **dorsal root ganglion (DRG)** in patients with history of varicella (chicken pox) infection.
   - Virus reproduce causing irritation of pain afferents in DRG leading to severe pain felt in dermatomal segment supplied by the infected ganglion.
   - virus also migrates with neuronal cytoplasmic flow to cutaneous terminals → painful skin rash and vesicular formation.

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➢ **Referred Pain:** Pain felt in a skin area originating from the same **embryonic** segment or dermatome as the diseased viscus (or deep structure) = supplied by the same dorsal roots. (**Dermatomal rule**). **Mechanisms:**

Convergence projection theory	Facilitation theory
• fibers of skin and viscus **converge** on same (**SGR**): finally activate same neurons • cortex **project** pain to a skin = **mis-interpretation caused by:** a- Brain is **more accustomed** to receive pain from skin than from viscera. b- Skin is **rich** in pain receptors and well **represented** in cortex. c- Skin is more **exposed** to stimulation	• fibers of diseased viscera, give subliminal fringe to nearby afferents from skin increasing their excitability, i.e. facilitating • Minor stimuli from skin can produce pain or hyperalgesia.

❖ **Examples:**

1- **Cardiac pain:** retrosternal, root of the neck and inner part of the left arm and also in the epigastrium.
2- **Gastric pain:** between the umbilicus and xiphoid process.
3- **Gall bladder pain:** in the mid-epigastrium and at tip of right scapula
4- **Renal pain:** as a back pain radiating to inguinal region and testicles in males
5- **Appendicitis pain:** around the umbilicus.
6- **Headache**: pain referred to surface of head from deep structures. **Types:**
   1) **Frontal:** Stimulation of pain receptors in areas supplied by trigeminal nerve
   2) **Occipital:** Stimulation of pain receptors in areas supplied by 2nd cervical

Headache of intracranial origins	Headache of extracranial origins
1- **Rise of intracranial tension: hemorrhage** or **mass.** 2- **drop of intracranial tension:** e.g. post lumbar- puncture 3- **Meningitis** or meningeal **irritation. (Alcohol)** 4- **Hypertension:** marked expansion of cerebral blood vessels. 5- **Migraine**: unilateral resulting from abnormal vascular phenomena • **N.B.: intracranial pain sensitive areas**: venous sinuses, tentorium, dura and the dural arteries especially middle meningeal artery (brain itself is insensitive to pain).	1. **Tension-Type:** spasm of neck muscles. 2. **Eye disorders:** errors of refraction. 3. **Sinusitis:** nasal sinuses. 4. **Otitis media:** inflammation of ear. 5. **Toothache.** 6. **Systemic disorders:** e.g. anemia

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Pain Control

- inhibiting Pain transmission in the central sensory pathways by ascending or descending impulses.
- **Suggesting observations include:**

a. degree that a person reacts to pain varies widely.
b. Injuries caused in stressful conditions as accidents or battles are ignored.
c. Irritation of the skin overlying a diseased viscus relieves pain.

❖ **The Gate Theory:**

- dorsal horn cells of the spinal cord, in particular SGR cells, act as a gate for transmission of pain sensation to the brain.
- Impulses coming along C pain fibers cause the release of substance P.
- can be partly or completely **closed by Presynaptic Inhibition → prevents release of substance P:**

(1) Opioid Peptides:

- Morphine is a powerful (analgesic) pain killer found in opium acts by combining with receptors in the CNS.
- **Types of endogenous opioid peptides:** Among about **12 opioid peptides** found in mammals, **three** are more important**:**

	Enkephalins	**Endorphins**	**Dynorphin**
Nature	Pentapeptides from pro-enkephalin	*B*-Endorphin is a 31 amino acid polypeptide derived from pro-opio-melano-cortin.	17 aa from prodynorphin
Site of release and function	a. areas of CNS, both supraspinal (periaqueductal grey area) and spinal (SGR): **analgesic effects**. b. limbic system where they promote a **sense of well being**	a. **Neurotransmitter:** hypothalamus project to thalamus and periaqueduct grey area b. **Neurohormone:** from anterior pituitary gland in stress into circulation: **Stress Analgesia.**	different brain stem nuclei. It is related to addiction and tolerance
Receptors	**Delta**	**Meu**	**Kappe**

N.B.: Opiate Receptors: in areas concerned with pain sensation, e.g. Periaqueductal gray area (PGA), SGR, thalamus, hypothalamus

(2) Supraspinal Descending Analgesia System: three major components:

	1. Periaqueductal gray area (PGA)	2. Raphe magnus nucleus	3. Pain inhibitory Interneurons
site	**midbrain and upper pons**, surrounding the aqueduct of Sylvius	**upper medulla**	dorsal horn of the **spinal cord**
sensitive to	**β-endorphin:** a. chemical transmitter from hypothalamus b. neurohormone from pituitary through blood stream in stress	**enkephalins**	**serotonin**
secrete	**enkephalins** to raphe magnus nucleus	**serotonin** to dorsal horn **inhibitory interneurons**	**Enkephalins** causes **presynaptic inhibition of SGR, preventing** release of **substance P**

- **N.B.; PGA may be stimulated by** Collaterals from **ascending pain pathway** (Lateral spinothalamic tract): **noxious stimuli** are important for activation of descending pain control system, **i.e. pain suppresses pain**

(3) Peripheral gate inhibition:

- stimulation of **large Aβ** sensory fibers from **peripheral tactile receptors** suppress pain transmission:
- tactile fibers send collaterals to activate **inhibitory interneurons → GABA or Enkephalins → presynaptic inhibition** of pain fibers at the SGR → closing the gate.
- **Examples:**

  a. Rubbing or massage of skin inhibits pain. b. Counter irritation, e.g. mustard plaster c. Acupuncture.

➢ **Methods of Treatment of Pain:**

Medical	Electric	Surgical (In case of uncontrollable severe pain)
• **Analgesics:** pain killers: as NSAID. • **Treatment of cause,** eg. Antiacids, antispasmodics or vasodilators	to analgesia system or to large sensory Aβ fibers	• **Antero-lateral cordotomy:** cut spinothalamic tract • **Prefrontal lobectomy**: relieves the emotional reaction to pain in cancer patients

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Lesions of the Sensory System

➢ Hyperalgesia:

- Abnormal increase in sensitivity to pain leading toExaggerated pain response
- **Importance**: self preservation mechanism: remind brain to avoid further injury
- **Types:**

	Primary hyperalgesia	**Secondary hyperalgesia**
occurs in	injured or inflamed skin area	intact skin
caused by	sensitization of peripheral nociceptors by inflammatory mediators	sensitization of central neurons of pain
threshold of pain	lowered	increased
		• pain produced is prolonged, severe and intolerable. • may be accompanied by **allodynia** (exaggerated pain response to non painful stimuli).
e.g.	**sunburned skin**	**Thalamic pain**

➢ Lesions of sensory pathway:

I- Peripheral nerve lesions:

1. **Mononeuropathy**: lesion of **one** peripheral nerve leads to loss of all sensations in area supplied by this nerve.
2. **Polyneuropathy (peripheral neuritis):**
   a. Diffuse lesion of **all peripheral** nerves, as in case of Vitamin B_{12} deficiency or diabetic neuritis,
   b. Affects peripheral at distal parts of the limbs leading to "**glove and stocking**" sensory disturbance (**numbness, tingling, burning** then total sensory loss).

N.B.: Injury / damage to a peripheral nerve can lead to chronic **neuropathic pain**

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II- Spinal cord Lesions:

1. Dorsal column lesion: Tabes Dorsalis:

- neurosyphilitic disease causes slow degeneration of sensory nerves in the dorsal root and dorsal column
- mostly in the lumbosacral spinal cord
- **leading to:**

a. Irritation of pain fibers and attacks of **severe pain**, **followed by:**

b. Degeneration and **Atrophy of Dorsal Column leading to:**

1) Loss of vibration sense.
2) Loss of proprioceptive sensation leads to **" Sensory Ataxia".**

NB: Ataxia: Incoordination of voluntary movements without paralysis. **Types:**

a. **Sensory ataxia:**
   - Due to loss of position sensation as in tabes dorsalis or pernicious anemia which cause destruction of posterior column
   - characterized by high steppage gait and +ve Romberg's sign (patient can't maintain erect position with closed eyes).

b. **Cerebellar ataxia:** due to lesion in neocerebellum.

c. **Vestibular ataxia:** due to lesion in the Vestibular division of the 8th nerve.

2. Central cord lesion: Syringomyelia:

- **widening of the central canal** by a fluid-filled cavity (or syrinx):
  damage of pain and temperature fibers crossing immediately in front of central canal first. Later, crude touch is affected.
- lesion is bilateral and usually affects the **cervical region** leading to **"jacket" loss of pain and temperature** sensations.
- Dorsal column sensations remain intact **(fine touch is intact**) and the condition is described as **dissociated sensory loss**

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3. Spinal cord Hemisection: (Brown Sequard syndrome):

		Sensory	Motor
At the level of lesion	same side	Loss of all sensations at the dermatome supplied by that posterior root entering at the site of injury	Flaccid paralysis (lower motor neuron lesion) and loss of all reflexes
Below the level of lesion	same side	Lesion to the dorsal column leads to loss of all sensations carried by it (fine touch, pressure, vibration, sense of position and of movement).	Lesion to corticospinal tract leads to spastic paralysis (upper motor neuron lesion), hyper-reflexia and +ve Babiniski
	opposite side	loss of pain and temperature (Lesion to the crossed spinothalamic tract), touch is not lost but decreased on both sides	no effect

III. Brain stem lesion: Rare (in brain stem strokes)

Ipsilateral sensory loss in face (supplied by trigeminal) with contralateral sensory loss below

IV. Thalamic Pain Syndrome: (Dejerine Roussy syndrome) = central post stroke pain (CPSP): rare condition after infarction of ventro-postero-lateral thalamus. **Patient presents with:**

1. Contralateral sensory loss.
2. Hemi-ataxia.
3. Thalamic pain: spontaneous bursts of intolerable treatment resistant pain (accompanied by 2ry hyperalgesia and allodynia

V- Cortical Lesions: in the somatic sensory cortex: sensations are not abolished. However, there is **loss of the following** from the **opposite side** of the body:

a- Fine touch, Stereognosis and Proprioceptive sensations.
b- Discrimination of mild grades of weights and temperature.
c- Localization of the source of sensations.
d- Pain localization, but pain sensation is least affected.

a- **Astereognosis:** patient loses ability to correlate shape, size, texture and weight of objects with previous experience.
b- **Amorphosynthesis** (Neglect syndrome): The patient neglects the contralateral side of his body, i.e. ignores its presence.